Multiplication & Division

Exam code: 0607

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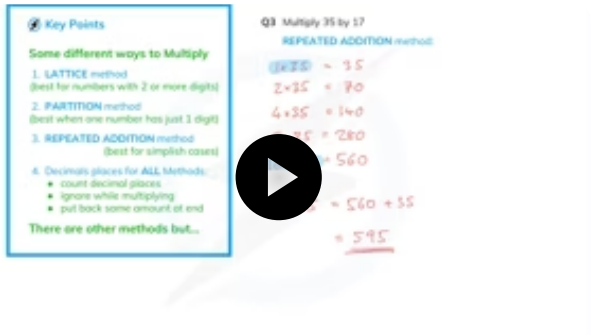

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Multiplication

How do I multiply two numbers without a calculator?

- There are a variety of written methods that can be used to add large numbers
  - You only need to know **one** method, but be able to use it confidently
  - Four common methods are described below, but there are many other valid methods

How do I use the column method?

- This is an efficient method if you are confident with multiplication
- To use the **column method**:
  - Write one number above the other lining up the digits using **place value columns**
  - Multiply the first digit (on the right) from the bottom value by each digit in the top value
    - Write the result under the line with the digits in the correct place value columns
  - Multiply the next digit in the bottom value by each digit in the top value
    - Always work from **right to left**
    - Use 0s as place holders when multiplying digits in columns other than the ones column

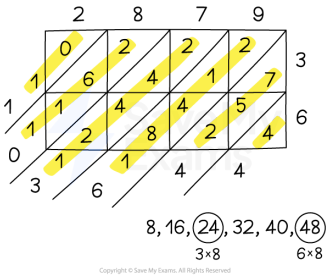
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$
\begin{array}{r}
522 \\
1740 \\
34800 \\
\hline
37062
\end{array}
$$

How do I use the lattice method?

- The lattice method is good for numbers with two or more digits
  - This method allows you to work with individual digits
- To use the **lattice method**:
  - Draw a grid
    - The **number of rows** should be the same as the number of digits in one number
    - The **number of columns** should be the same as the number of digits in the other number
    - Draw diagonal lines through the boxes
  - Multiply each pair of digits, writing the result in the relevant box
    - Ones should be written in the bottom half of the box and tens in the top half of the box
  - Add the digits along the diagonals and write the result in the diagonal outside the grid
    - Carry the tens of any 2 digit result into the next diagonal
- For example, 2879 × 36 = 103 644



How do I use the grid method?

- This method keeps the value of the larger number intact
  - It may take longer with two larger numbers
  - Be careful lining up numbers with lots of zeros!
- To use the **grid method**
  - Draw a grid
    - The **number of rows** should be the same as the number of digits in one number
    - The **number of columns** should be the same as the number of digits in the other number
  - Label the rows and columns with the values of each digit
    - E.g. For 3516 you would use 3000, 500, 10 and 6
  - Multiply together the relevant values and put the results in the boxes

	3000	500	10	6
7	21 000	3500	70	42

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How do I use the repeated addition method?

- This is best for smaller, simpler cases
  - You may have seen this called 'chunking'
- To use the **repeated addition method**
  - Build up to the answer using simple multiplication facts that can be worked out easily
  - To find 13×23 :
    $1 \times 23 = 23$
    $2 \times 23 = 46$
    $4 \times 23 = 92$
    $8 \times 23 = 184$
  - So, $13 \times 23 = 1 \times 23 + 4 \times 23 + 8 \times 23 = 23 + 92 + 184 = 299$

What words are used for multiplication and division?

- **Multiplication** may be phrased using the words **lots of, times** or **product**
- **Division** may be phrased using the words **quotient, share** and **per**

Examiner Tips and Tricks

A good way to check your answer without a calculator is to estimate it (e.g. by rounding everything to 1 significant figure).

Worked Example

Multiply 725 by 19.

Answer:

Using the grid method

- Split the numbers up using the place value
- Multiply numbers to fill in the grid

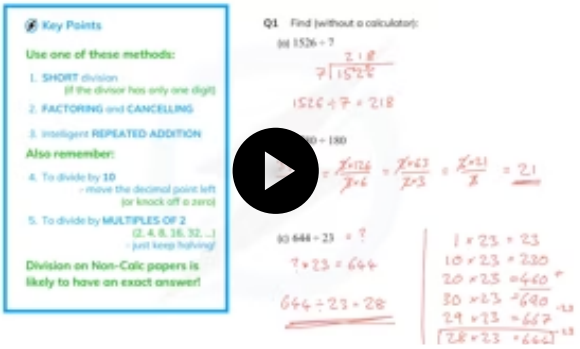
×	700	20	5
10	7000	200	50
9	6300	180	45

$$
\begin{array}{r}
50 \\
6300 \\
180 \\
+\quad 45 \\
\hline
13775
\end{array}
$$

Use an estimate to check your answer

- 700×20 is equal to 14 000

725 × 19 = 13 775



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Division

How do I divide a number by another without a calculator?

- The most common written method for division is short division (or "the bus stop method")
  - There are other methods such as long division, but short division is generally the most efficient
- While short division is best when dividing by a single digit, for bigger numbers you may need a different approach
- You can use other number skills to help
  - eg. cancelling fractions, “shortcuts” for dividing by 2 and 10, and the repeated addition (“chunking”) method covered in Multiplication

Short division (bus stop method)

- Unless you can use simple shortcuts such as dividing by 2 or by 10, this method is best used when dividing by a single digit
- To find 174 ÷ 3
  - Starting from the left; 3 fits into 1, 0 times, with a remainder of 1
    - Carry the remainder of 1 over to the next digit, which forms 17
    - $\begin{array}{r} 0 \\ 3\overline{)1\,{}^{1}7\;\;4} \end{array}$

- $3\overline{)1\,{}^{1}7\,{}^{2}4}$ $\;0\;5$

- 3 fits into 24, 8 times, with no remainder
  - $3\overline{)1\,{}^{1}7\,{}^{2}4}$ $\;0\;5\;8$
- So, 174 ÷ 3 = 58

Factorising & cancelling

- This involves treating division as you would if you were asked to simplify fractions
  - For example, 1008 ÷ 28 can be written as $\frac{1008}{28}$
  - This can then be simplified
    - $\frac{1008}{28} = \frac{504}{14} = \frac{252}{7}$
    - 252 ÷ 7 can then be calculated using short division; the answer is 36

Dividing by 10, 100, 1000, … (Powers of 10)

- This is a case of shifting digits along the place value columns
  For example
  - 380 ÷ 10 = 38.0 (shifts by 1 column)
  - 45 ÷ 100 = 0.45 (shifts by 2 columns)
  - 28 ÷ 1000 = 0.028 (shifts by 3 columns)
    - For cases like this, it can help to add leading zeros
    - 0028 ÷ 1000 may be easier to visualise

Worked Example

Divide 568 by 8.

Answer:

This is division by a single digit so short division would be an appropriate method
If you spot it though, 8 is also a power of 2 so you could just halve three times

Using short division, the bus stop method:

8 fits into 5, 0 times, with a remainder of 5
8 fits into 56, 7 times exactly
8 fits into 8, 1 time exactly

Use an estimate to check your answer

- 600 ÷ 10 is equal to 60

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